Oracle Utilities Service and Measurement Data Foundation v25.10HOT

Product Fix Design

Bug 38966529

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Oracle Utilities Service and Measurement Data Foundation Product Fix Design

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Table of Contents

Reported Issue **4**
Description 4
Steps To Recreate 4

Solution Design **5**
Summary 5
Other Changes 5
Algorithms 5

Data Clean-Up / Fixes **5**

Recommended Test Cases **6**
Test Case 1: subUT created when period is 24 hours due to rate change 6
Test Case 2: Verify that sub usage transaction does not throw any error and does not have scalar usage period details when the scalar period is less than a day. 6
Test Case 3: No subUT created when period is less tha 24 hours due to rate change 7

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Reported Issue

Description

Rate Change in middle of period (Rate change a day after bill segment start date). The regenerated bill segment comes with 1 bseg header only covering the new rate schedule (missing the old rate schedule).

When there is rate change on usage subscription that falls on the last day of billing, the usage transaction ends in issue for the last day as it is requiring a read to be present on the subUT created for the rate change.

Steps To Recreate

1. Ensure algorithm Derive Usage Group Based On Rate and Installed Device (D2-DRVUSGGRP) has Suppress One Day Effective Rates (Y/N) = Y
2. Create usage subscription
    - Ensure rate change happens on 2-1-2025
    - Ensure interval and scalar present on device
        - Ensure interval MC has reads from 01-01-2025 12:00AM to 02-02-2025 12:00AM
        - Ensure scalar MC has reads for 01-01-2025 12:00AM and 02-01-2025 12:00AM
3. Create usage transaction for dates
    - Interval Start 01-01-2025 12:00AM
    - Interval End From 02-02-2025 12:00AM
    - Interval End To 02-02-2025 12:00AM
    - Scalar Start 01-01-2025 12:00AM
    - Scalar End From 02-01-2025 12:00AM
    - Scalar End To 02-01-2025 11:59AM

Verify that UT is processed and subUT is created for 1 day period.

Solution Design

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Summary

When rate changes occur, UT processing creates subUTs for these date changes only when period is more than 1 day.

Modify the algorithm Derive Usage Group Based On Rate and Installed Device, to suppress creation of subUTs only when the rate changes results to subUTs with less than 24 hour span.

Change Type	Y/N
Does this fix require any Admin (Soft) Table Changes to be made by the client?	N
Does this fix require upgrade data changes to be applied?	N
Does the fix involve DB Changes?	N
Is there an impact on any user interface or any other on-line behavior?	Y
Is there an impact on any batch process?	N
Is there an impact on CM – Custom Modification processing?	N
Does this fix require an application viewer update for Algorithm data (run F1-AVALG)?	N
Does this fix require an application viewer update for MO data (run F1-AVMO)?	N
Does this fix require an application viewer update for Table data (run F1-AVTBL)?	N

Other Changes

Algorithms

Derive Usage Group Based On Rate and Installed Device (D2-DRVUSGGRP) (Change)

Modify the algorithm Derive Usage Group Based On Rate and Installed Device, to suppress creation of subUTs only when the rate changes results to subUTs with less than 24 hour span.

Data Clean-Up / Fixes

Not applicable

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Recommended Test Cases

Test Case 1: subUT created when period is 24 hours due to rate change

4. Ensure algorithm Derive Usage Group Based On Rate and Installed Device (D2-DRVUSGGRP) has Suppress One Day Effective Rates (Y/N) = Y
5. Create usage subscription
    - Ensure rate change happens on 2-1-2025
    - Ensure interval and scalar present on device
        - Ensure interval MC has reads from 01-01-2025 12:00AM to 02-02-2025 12:00AM
        - Ensure scalar MC has reads for 01-01-2025 12:00AM and 02-01-2025 12:00AM
6. Create usage transaction for dates
    - Interval Start 01-01-2025 12:00AM
    - Interval End From 02-02-2025 12:00AM
    - Interval End To 02-02-2025 12:00AM
    - Scalar Start 01-01-2025 12:00AM
    - Scalar End From 02-01-2025 12:00AM
    - Scalar End To 02-01-2025 11:59AM

Verify that UT is processed and subUT is created for 1 day period.

Test Case 2: Verify that sub usage transaction does not throw any error and does not have scalar usage period details when the scalar period is less than a day.

1. Create a device, device configuration, scalar and interval measuring component.
2. Create a service point and install the device on the service point.
3. Create measurements on the scalar measuring component from 01-01-2025 to 02-01-2025 12 A.M.

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4. Create measurements on the interval measuring component from 01-01-2025 to 02-05-2025 12A.M.
5. Create a usage calculation group with get scalar and get interval usage rules.
6. Add two rates in the D2-CCBRateScheduleLookup extendable lookup and configure the above created usage calculation group.
7. Create a usage subscription type with calculation group determination override algorithm - Derive Usage Group Based On Rate and Installed Device (D2-DRVUSGGRP) configured.
8. Create a usage subscription using the above type and the rate history for the following dates –
    a. 01-01-2025 – Rate 1
    b. 02-01-2025 – Rate 2
9. Create a usage transaction for the following dates –

    Interval Measuring component –

    Start Date/Time: 01-01-2025 12A.M

    End Date/Time From: 02-05-2025 12 A.M

    End Date/Time To: 02-05-2025 12A.M

    Scalar Measuring component –

    Start Date/Time: 01-01-2025 12A.M

    End Date/Time From: 02-01-2025 12 A.M

    End Date/Time To: 02-01-2025 11:59 P.M
10. Verify that usage transaction moves to approval in progress and two sub usage transactions are created, 01-01-2025 to 02-01-2025 and 02-01-2025 to 02-05-2025.

Verify that sub usage transaction with dates 02-01-2025 and 02-05-2025 is in calculated state with only interval usage period details. Verify that scalar usage is not present as start date/time – 02-01-2025 12 AM and scalar processing end date/time-from are same – 02-01-2025 12A.M (scalar period is less than one day).

Test Case 3: No subUT created when period is less tha 24 hours due to rate change

1. Ensure algorithm Derive Usage Group Based On Rate and Installed Device (D2-DRVUSGGRP) has Suppress One Day Effective Rates (Y/N) = Y
2. Create usage subscription
    - Ensure rate change happens on 01-31-2025

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- Ensure interval and scalar present on device
  - Ensure interval MC has reads from 01-01-2025 12:00AM to 02-02-2025 12:00AM
  - Ensure scalar MC has reads for 01-01-2025 12:00AM and 02-01-2025 12:00AM

3. Create usage transaction for dates
   - Interval Start 01-01-2025 12:00AM
   - Interval End From 01-31-2025 12:00PM
   - Interval End To 01-31-2025 12:00PM
   - Scalar Start 01-01-2025 12:00AM
   - Scalar End From 01-31-2025 12:00AM
   - Scalar End To 01-31-2025 11:59AM

Verify that UT is processed and No subUT is created for less than 1 day period.

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